

1. 2.10×10^3 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
2. 2.0×10^4 Pa. $\therefore$ the balanced equation determines the amount of CO_2 produced from the amount of HCl that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
3. 22 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:2 mole ratio, so the gas volumes must change in the same 1:2 ratio.
4. 344 K. $\therefore$ rearranging the combined gas law gives $T_2 = T_1(P_2V_2)/(P_1V_1)$. Absolute temperature is directly proportional to the product of pressure and volume for a fixed amount of gas, so kelvin must be used.
5. HBr . $\therefore$ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 80.91 g/mol. Only HBr has that molar mass, so it is the only identity consistent with the measurements.
6. 1.9×10^3 kPa. $\therefore$ the amount of gas and the temperature remain constant, the product PV remains constant. Reducing the volume causes molecules to collide with the container walls more frequently, increasing the pressure by the same factor that the volume decreases.
7. 72 N. The piston is rectangular, so its area is length $\times$ width. $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
8. 0.50. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
9. 0.56. $\therefore$ 20. g of ammonium hydrogen carbonate corresponds to 0.25 mol, and the reaction stoichiometry produces 0.25 mol of carbon dioxide. The argon remains at 0.20 mol because it does not participate in the reaction. The mole fraction is the ratio of the moles of the specified gas to the total moles of gas present after the reaction, so it depends only on the final amounts of each gas.
10. 0.82 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
11. 0.11 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
12. Remains unchanged. $\therefore$ the container volume is fixed and the amount of gas remains constant, both the mass and the volume stay constant. Heating changes the pressure, not the mass per unit volume.
13. 9.32 cm^3 . SINCE at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:2 ratio of H_2 to HBr , so the product volume is multiplied by 2/1.
14. 0.32. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
15. 2.78 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
16. 7.4 atm. $\therefore$ no gas is added or removed before equilibrium is reached, the amount of gas remains constant. The initial pressure equals the factor by which the gas expands when the final pressure is atmospheric pressure.
17. Xe . $\therefore$ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 131.29 g/mol. Only Xe has that molar mass, so it is the only identity consistent with the measurements.
18. 3.42×10^4 Pa. $\therefore$ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
19. 2.7×10^3 Pa. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
20. 11 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the $\text{N}_2\text{O}_4:\text{NO}_2$ coefficient ratio of 2:1 determines the required volume.
21. 1.8 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes

by the same ratio as the Kelvin temperatures.

22. 0.243 atm. $\therefore$ removing one gas leaves only the pressure generated by the remaining particles, and that pressure equals the remaining gas's mole fraction of the original total pressure.
23. 3.3 L. $\therefore$ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until $P_i V_i = P_f V_f$. Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
24. Yes; final temperature = 511.°C. $\therefore$ the amount of gas and volume remain constant, the ideal gas law requires pressure and absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.
25. 0.40 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
26. 0.14 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
27. Consistent. $\therefore$ Combining the measured density with the ideal gas law gives the experimental molar mass, which matches the molar mass of C_2H_6 .
28. 6.5 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 1:1 particle ratio between CO_2 and CS_2 , so their volumes must be in the same ratio.
29. $9.8 \times 10^4\text{ Pa}$. $\therefore$ the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.
30. 528 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
31. $2.8 \times 10^3\text{ Pa}$. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
32. 0.17 mol. $\therefore$ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by $PV = nRT$, giving $n = PV/RT$.
33. CO. $\therefore$ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 28.01 g/mol. Only CO has that molar mass, so it is the only identity consistent with the measurements.
34. 57 g/mol. $\therefore$ density gives the mass per unit volume, while the ideal gas law relates volume to the number of moles, combining these relationships yields the mass of one mole of the gas.
35. Remains unchanged. $\therefore$ the equilibrium vapor pressure of a pure substance depends only on temperature. Adding more liquid changes the amount of liquid present but does not change the equilibrium established between evaporation and condensation.
36. $4.1 \times 10^3\text{ Pa}$. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
37. 2.4 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
38. 38% of the initial volume. $\therefore$ rearranging the combined gas law gives $V_2/V_1 = (T_2/T_1)(P_1/P_2)$. The final volume is determined by the competing effects of the change in absolute temperature and the change in pressure.
39. $8.3 \times 10^3\text{ Pa}$. $\therefore$ the balanced equation determines the amount of CO_2 produced from the amount of $MgCO_3$ that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
40. 7.86 cm^3 . SINCE at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:3 ratio of N_2H_4 to N_2 , so the product volume is multiplied by 3/2.
41. 0.71. $\therefore$ 21.0 g of iron corresponds to 0.376 mol, and the reaction stoichiometry produces 0.38 mol of hydrogen. The argon remains at 0.15 mol because it does not participate in the reaction. The mole fraction is the ratio of the moles of the specified gas to the total moles of gas present after the reaction, so it depends only on the final amounts of each gas.
42. 0.556 atm. $\therefore$ each gas contributes the same fraction of the total pressure as its fraction of the total number of moles, the mole ratio determines the pressure ratio.

43. 6.1 mL. $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{O}_2:\text{H}_2\text{O}$ is 3:2 so the volume changes by the same factor.
44. 86 g/mol. $\therefore$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
45. 57 g/mol. $\therefore$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
46. 14 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 4:6 mole ratio, so the gas volumes must change in the same 4:6 ratio.
47. 2.8 L. $\therefore$ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
48. 1.9×10^4 Pa. $\therefore$ the balanced equation determines the amount of H_2 produced from the amount of Mg that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
49. 71.9 N Only the piston area is needed because pressure is force per unit area; the gas height does not affect the force once the pressure is known. $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
50. 54 g/mol. $\therefore$ the measured pressure, volume, and temperature determine the number of moles through the ideal gas law, dividing the measured mass by that amount gives the molar mass.
51. 5.5 cm^3 . SINCE at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:1 ratio of HF to H_2SiF_6 , so the product volume is multiplied by 1/2.
52. 29.8 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:2 mole ratio, so the gas volumes must change in the same 1:2 ratio.
53. 2.79×10^3 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
54. Increases. $\therefore$ increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.
55. 136 N The piston is circular, so its area is $\pi(d/2)^2$. $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
56. Yes; final temperature = $-194.^\circ\text{C}$. $\therefore$ the amount of gas and volume remain constant, the ideal gas law requires pressure and absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.
57. Decreases. $\therefore$ the molar concentration is unchanged at the same temperature and pressure, but each mole has less mass. Less mass therefore occupies the same volume.
58. 0.832. $\therefore$ removing one gas leaves only the pressure produced by the remaining gas, and pressure is proportional to the number of moles at constant temperature and volume.
59. 9.64×10^4 Pa $\therefore$ the total pressure equals the sum of the partial pressures of all gases present, including the water vapor, before converting to SI units.
60. 9.818×10^4 Pa $\therefore$ a barometer measures atmospheric pressure directly, so the reading is simply converted from torr to pascals.
61. 0.260. $\therefore$ 11.3 g of lithium carbonate corresponds to 0.153 mol, and the reaction stoichiometry produces 0.153 mol of carbon dioxide. The argon remains at 0.436 mol because it does not participate in the reaction. The mole fraction is the ratio of the moles of the specified gas to the total moles of gas present after the reaction, so it depends only on the final amounts of each gas.
62. 9.56×10^4 Pa $\therefore$ the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
63. 9.82×10^4 Pa $\therefore$ the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
64. 0.17. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
65. 5.0 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation

gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the NO₂:O₂ coefficient ratio of 0.5:1 determines the required volume.

66. 2.7×10^4 Pa. $\therefore$ pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
67. 4.3 cm^3 . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the NO₂:NO coefficient ratio of 1:1 determines the required volume.
68. 5.6 L. $\therefore$ the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product PV remains unchanged.
69. 7.1×10^5 Pa. $\therefore$ rearranging the combined gas law gives $P_1 = P_2(V_2/V_1)(T_1/T_2)$. Pressure is directly proportional to absolute temperature and inversely proportional to volume for a fixed amount of gas, so both changes must be accounted for simultaneously.
70. $10. \text{ cm}^3$. At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the H₂:H₂O coefficient ratio of 1:1 determines the required volume.
71. Increases. $\therefore$ adding gas increases the mass while the container volume does not change. More mass per unit volume produces a greater density.
72. 0.17. $\therefore$ 19 g of sodium peroxide corresponds to 0.24 mol, and the reaction stoichiometry produces 0.12 mol of oxygen. The nitrogen remains at 0.590 mol because it does not participate in the reaction. The mole fraction is the ratio of the moles of the specified gas to the total moles of gas present after the reaction, so it depends only on the final amounts of each gas.
73. 19 g/mol. $\therefore$ the measured pressure, volume, and temperature determine the number of moles through the ideal gas law, dividing the measured mass by that amount gives the molar mass.
74. 8.29 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
75. 1.1×10^4 Pa. $\therefore$ a closed-end manometer has a vacuum at one end, so the height difference equals the absolute gas pressure before converting to SI units.
76. Increases. $\therefore$ increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.
77. 0.191 m. $\therefore$ Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through $A = F/P$. Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
78. 5.07×10^4 Pa. $\therefore$ the mole fractions must sum to 1, so the mole fraction of N₂ is the remainder. Dalton's law states that a gas's partial pressure equals its mole fraction multiplied by the total pressure.
79. 18 mL. $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for CO₂:HCl is 1:2 so the volume changes by the same factor.
80. 5.59 L. $\therefore$ the ideal gas law gives $PV = nRT$, but the amount of gas is not given directly. The amount must first be determined from the sample mass using $n = m/M$, after which solving $PV = nRT$ for volume yields the correct result.
81. 34 N. The piston is square, so its area is side². $\therefore$ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
82. 1.00×10^5 Pa. $\therefore$ a barometer measures atmospheric pressure directly, so the reading is simply converted from torr to pascals.
83. 25.4 cm^3 . SINCE at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:2 ratio of CS₂ to SO₂, so the product volume is multiplied by 2/1.
84. 0.36. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
85. 2.09×10^4 Pa. $\therefore$ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
86. Not consistent. $\therefore$ Combining the measured density with the ideal gas law gives the experimental molar mass, which does not match the molar mass of HBr.
87. 4.0 L. $\therefore$ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until $P_i V_i = P_f V_f$. Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.

88. 0.339 atm. Pressure is the same physical quantity regardless of the unit used. The numerical value changes only because 1 atm corresponds to 101.325 kPa, so expressing the same pressure in larger units gives a smaller number.
89. 2.45×10^3 mol. $\therefore$ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n .
90. No. $\therefore$ keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
91. 110 g/mol. $\therefore$ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
92. 3.2 cm^3 . $\therefore$ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:1 particle ratio between NO_2 and O_2 , so their volumes must be in the same ratio.
93. 30. g/mol. $\therefore$ the measured pressure, volume, and temperature determine the number of moles through the ideal gas law, dividing the measured mass by that amount gives the molar mass.
94. 13 mL. $\therefore$ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:1 mole ratio, so the gas volumes must change in the same 1:1 ratio.
95. Increases. $\therefore$ compression reduces the gas volume without changing its mass. The same mass occupying a smaller volume results in a greater density.
96. 1.74 mL. $\therefore$ at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for $\text{NH}_3:\text{N}_2$ is 2:1 so the volume changes by the same factor.
97. 31 cm. $\therefore$ the total amount of gas is unchanged, the initial cylinder volume is obtained from the pressure ratio. The cylinder height then follows from the volume of a cylinder.
98. $1.3 \times 10^4 \text{ Pa}$ $\therefore$ a closed-end manometer has a vacuum at one end, so the height difference equals the absolute gas pressure before converting to SI units.
99. 2.8 L. $\therefore$ the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product PV remains unchanged.
100. $9.5 \times 10^4 \text{ Pa}$ $\therefore$ the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.